

Properties of trapezoids



1

a $\overline{AD}$ and $\overline{BC}$

c $m\angle ADC = 131^\circ$

b $m\angle ABC = 60^\circ$

2

a $z = 26^\circ$

c $x = 114, y = 114$

b $x = 85^\circ, y = 74^\circ$

d $x = 90, y = 110$

3

a $m\angle B = 128^\circ$

c $m\angle D = 52^\circ$

b $m\angle C = 128^\circ$

4 $JL = 20$

5 $x = 13$

6 5

7

a $x = 69$

b $y = 2$

8

a $x = 12$

b 50°

9

a Yes

c No

b No

d No

10

a No

c Yes

b No

d No

11



To prove: Each pair of base angles of trapezoid $ABCD$ are congruent.

Statements	Reasons
1. $ABCD$ is an isosceles trapezoid	Given
2. $\overline{AD} \cong \overline{BC}$	Definition of an isosceles trapezoid
3. $\overline{AE} \cong \overline{BF}$	Altitudes of a trapezoid are congruent
4. $\angle DEA$ and $\angle CFB$ are right angles	Definition of an altitude
5. $\angle DEA \cong \angle CFB$	All right angles are congruent
6. $\triangle AED \cong \triangle BFC$	Hypotenuse-leg congruence theorem
7. $\angle ADE \cong \angle BCF$	Corresponding parts of congruent triangles are congruent (CPCTC)
8. $\angle BCD$ and $\angle CBA$ are supplementary	Consecutive interior angles postulate
9. $\angle ADC$ and $\angle BAD$ are supplementary	Consecutive interior angles postulate
10. $\angle BAD \cong \angle ABC$	Congruent supplements theorem

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To prove: $\overline{AC} \cong \overline{BD}$

Statements	Reasons
1. $ABCD$ is an isosceles trapezoid	Given
2. $\overline{AD} \cong \overline{BC}$	Definition of an isosceles trapezoid
3. $\overline{DC} \cong \overline{DC}$	Reflexive property of congruence
4. $\angle ADC \cong \angle BCD$	If a trapezoid is isosceles, then each pair of base angles is congruent.
5. $\triangle ADC \cong \triangle BCD$	Side-angle-side congruence theorem
6. $\overline{AC} \cong \overline{BD}$	Corresponding parts of congruent triangles are congruent (CPCTC)



Additional questions

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a $m\angle 1 = 77^\circ$
 $m\angle 2 = 108^\circ$
 $m\angle 3 = 108^\circ$

c $m\angle 1 = 120^\circ$
 $m\angle 2 = 120^\circ$
 $m\angle 3 = 60^\circ$

b $m\angle 1 = 69^\circ$
 $m\angle 2 = 69^\circ$
 $m\angle 3 = 111^\circ$

d $m\angle 1 = 105^\circ$
 $m\angle 2 = 75^\circ$
 $m\angle 3 = 75^\circ$

14 $FD = 3.5$

15 $FD = 1.26$

16 $x = 5^\circ$

17 $x = 7^\circ$

18 $x = 6$

19

a $x = 3$

c $TU = 19.5$

b $BC = 27$

d $KH = 12$

20 $x = 4$

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- a True, since the legs of an isosceles trapezoid are congruent.
- b True, since the diagonals of an isosceles trapezoid are congruent.
- c False, the two bases of a trapezoid are not congruent.
- d False, the legs of a trapezoid are not parallel.
- e False, opposite angles in a trapezoid are not congruent.
- f True, since the adjacent angles on the same leg of an isosceles trapezoid are congruent.

To prove:  $\cong \angle D$

Statements	Reasons
1. Trapezoid $ABCD$ with altitudes $\overline{BE}$ and $\overline{CF}$	Given
2. $\overline{AB} \cong \overline{CD}$	Given
3. $\overline{BE} \cong \overline{CF}$	Definition of altitude
4. $\angle AEB, \angle CFD$ are right angles	Definition of altitude
5. $\triangle AEB, \triangle CFD$ are right triangles	Definition of right triangle
6. $\triangle AEB \cong \triangle CFD$ are right triangles	HL postulate
7. $\angle A \cong \angle D$	CPCTC

To prove: Trapezoid $ABCD$ is isosceles

Statements	Reasons
1. Trapezoid $ABCD$	Given
2. $\overline{EF} \cong \overline{EG}$	Given
3. $\overline{AF} \cong \overline{DG}$	Given
4. $\overline{BG} \cong \overline{CF}$	Given
5. $AF + FG = AG$	Segment addition postulate
6. $DG + FG = FD$	Segment addition postulate
7. $AF + FG = FD$	Substitution property
8. $AG = FD$	Substitution property
9. $\overline{AG} \cong \overline{FD}$	Definition of congruence
10. $\triangle FEG$ is isosceles	Definition of isosceles triangle
11. $\angle EFG \cong \angle EGF$	Definition of isosceles triangle
12. $\triangle BAG \cong \triangle CDF$	SAS congruence theorem
13. $\overline{BA} \cong \overline{CD}$	CPCTC
14. Trapezoid $ABCD$ is isosceles	Definition of isosceles trapezoid

Statements	Reasons
1. Trapezoid $ABCD$	Given
2. $\angle BAK \cong \angle BKA$	Given
3. $\overline{KD} \parallel \overline{BC}$	Definition of isosceles trapezoid
4. $\angle BKA \cong \angle KBC$	Converse of the alternate interior angles theorem
5. $\angle CDK \cong \angle BAK$	Definition of isosceles trapezoid
6. $\angle KBC \cong \angle CDK$	Transitive property
7. $\angle KBC$ and $\angle BKD$ are supplementary angles	Converse of the consecutive interior angles theorem
8. $\angle KBC + \angle BKD = 180^\circ$	Definition of supplementary angles
9. $\angle BCD$ and $\angle CDK$ are supplementary angles	Converse of the consecutive interior angles theorem
10. $\angle BCD + \angle CDK = 180^\circ$	Definition of supplementary angles
11. $\angle KBC + \angle BKD = \angle BCD + \angle CDK$	Substitution property
12. $\angle CDK + \angle BKD = \angle BCD + \angle CDK$	Substitution property
13. $\angle BKD = \angle BCD$	Subtraction property
14. $\angle BKD \cong \angle BCD$	Definition of congruence
15. $BKCD$ is a parallelogram	Definition of parallelogram